

AlphaLeak

AlphaLeak: What ‘Blind’ LLM Trading Benchmarks Still See

Masking company names does not make an LLM trading benchmark blind. Masked records are traced back to their stock 99.8% of the time, and the formats that truly hide identity add no measurable ranking value.

99.8%

name-masked records re-identified
strongest of 5 retrieval attacks; chance 1.37%

71%

issuers named by an LLM with no candidate list
95% interval [0.61, 0.80]

5.1% / 3.9%

still re-identified under Theme / Event tags
but their ranking gain is not distinguishable from zero

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Hiding the name does not hide the company

A one-minute read for someone new to the topic. Slides 03–09 give the evidence.

In plain terms

The problem Benchmarks that test whether an LLM can trade often "blind" it by replacing company names with codes, assuming it can no longer tell which company it is reading about. But the article still says GPUs, Santa Clara and export curbs.

The idea Audit every masking format two ways on the same records: can an attacker still find the stock (linkability), and does the text still help the model rank stocks (utility)? Seven formats, five retrieval attacks, two LLM adversaries.

The result Name-masked records were traced back to their stock 99.8% of the time (chance: 1.37%). The only formats that truly hide identity, theme and event tags, add no measurable ranking value; the apparent edge of raw news is a sector tilt, not stock picking.

MY ROLE · WHERE THIS COULD BE USED

My role Third author.

Use case Validating an LLM trading backtest or benchmark before trusting it; deciding what a data release may safely contain.

Terms used in this deck

RE-IDENTIFICATION (RECOVERY) RATE

How often an attacker finds the true stock from a masked record; 1.37% would mean the mask works.

RANK IC

Correlation between the model's stock ranking and what happened next; 0 means no better than ignoring the news.

BLIND vs MASKED

Masking is what was done to the text; blindness is whether identity actually survived an attack.

PLACEBO ARTICLE

A swapped-in article about another stock, used to separate content from identity.

SECTOR TILT

Returns that come from favouring whole sectors rather than from picking individual stocks.

A mask is not a blindfold

“Blind” benchmarks swap tickers for codes and assume the model can no longer tell which company it is reading about.

Why it matters

- NVIDIA becomes STOCK_042, but the text still mentions GPUs, Santa Clara and export curbs.
- If the record reveals the issuer, a benchmark score mixes event reasoning with memorized issuer and sector priors.
- Masking is an operation. Blindness is a property of the whole released record, including numbers, pseudonyms and repeated scores.
- Portfolio returns can turn sector tilts into what looks like stock-picking skill.

WHAT THIS MEANS OUTSIDE THE PAPER

Blindness is measured a record must survive an attack, not merely lose its names.

Both sides audited how findable the subject is, and whether the text still helps.

Generalizes to any evaluation set that claims anonymity after redaction.

Data & setup

PANEL A

100 S&P 500 stocks × 17 monthly rebalances, Jun 2023–Oct 2024, scored by GPT-5.5

PANEL B

200 stocks × 24 dates × 4 LLMs: Claude, Gemini, GPT-5.5, Grok

INPUTS

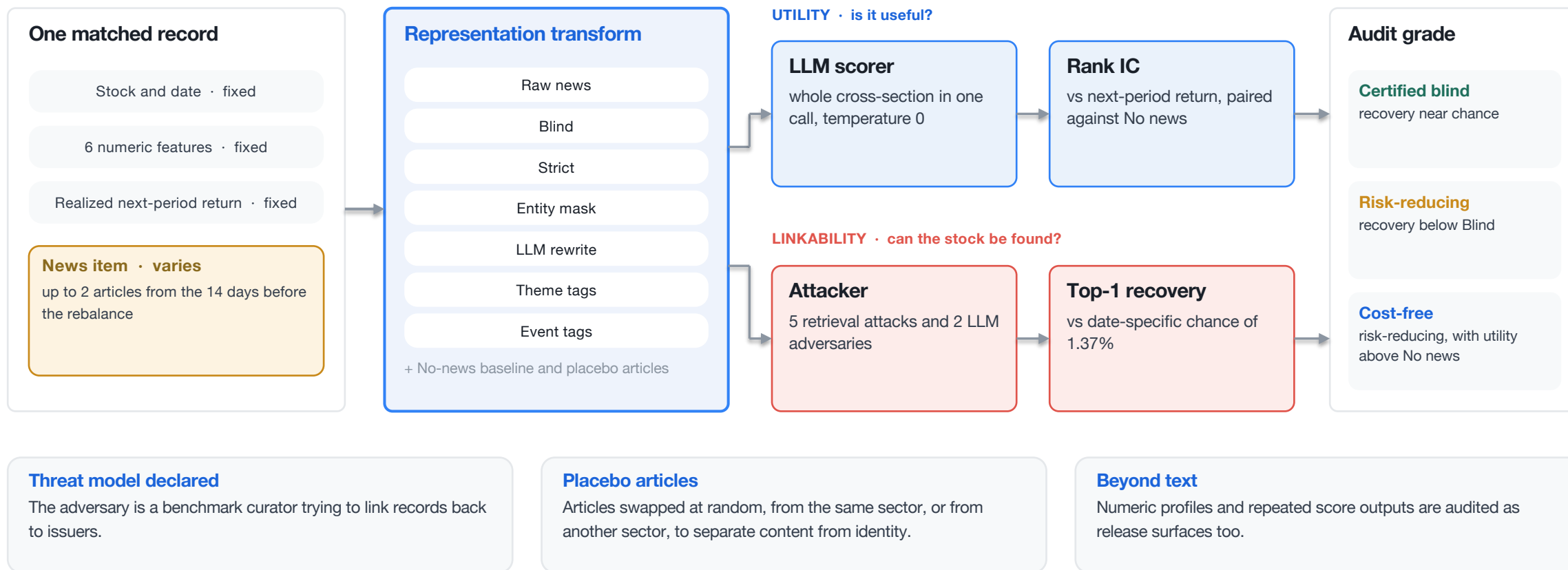
6 numeric fields plus up to 2 time-stamped articles per stock

SCORING

Whole cross-section in one call; scores in $[-1, 1]$ at temperature 0

A matched audit: one record, many masks, two tests

Everything is fixed except what the model sees; each representation is scored for usefulness and for identifiability.



Redrawn and simplified from the paper's Figure 1.

The same news item under seven representations

Red text is an identity cue that survived; the bar shows how often the strongest of five attacks still finds the stock.

REPRESENTATION	NEWS FIELD SHOWN TO THE LLM	ATTACKS STILL FIND THE STOCK
Raw news unredacted news	NVIDIA beats on GPU demand; Santa Clara export curbs	1.000
Blind name replaced by a label	STOCK_042 beats on GPU demand; Santa Clara export curbs	0.998
Strict one blanket placeholder	STOCK_042 beats on MASKED_ENTITY demand; MASKED_ENTITY export curbs	0.951
Entity mask same spans, type kept	STOCK_042 beats on MASKED_TICKER demand; MASKED_NAME export curbs	0.951
LLM rewrite LLM rewrite, event kept	Large-cap chipmaker beats on compute demand ; export-curb risk	0.918
Theme tags keyword tags	earnings guidance regulation	0.051
Event tags tags + actions + tone	earnings beat guidance raise reg. risk tone +	0.039

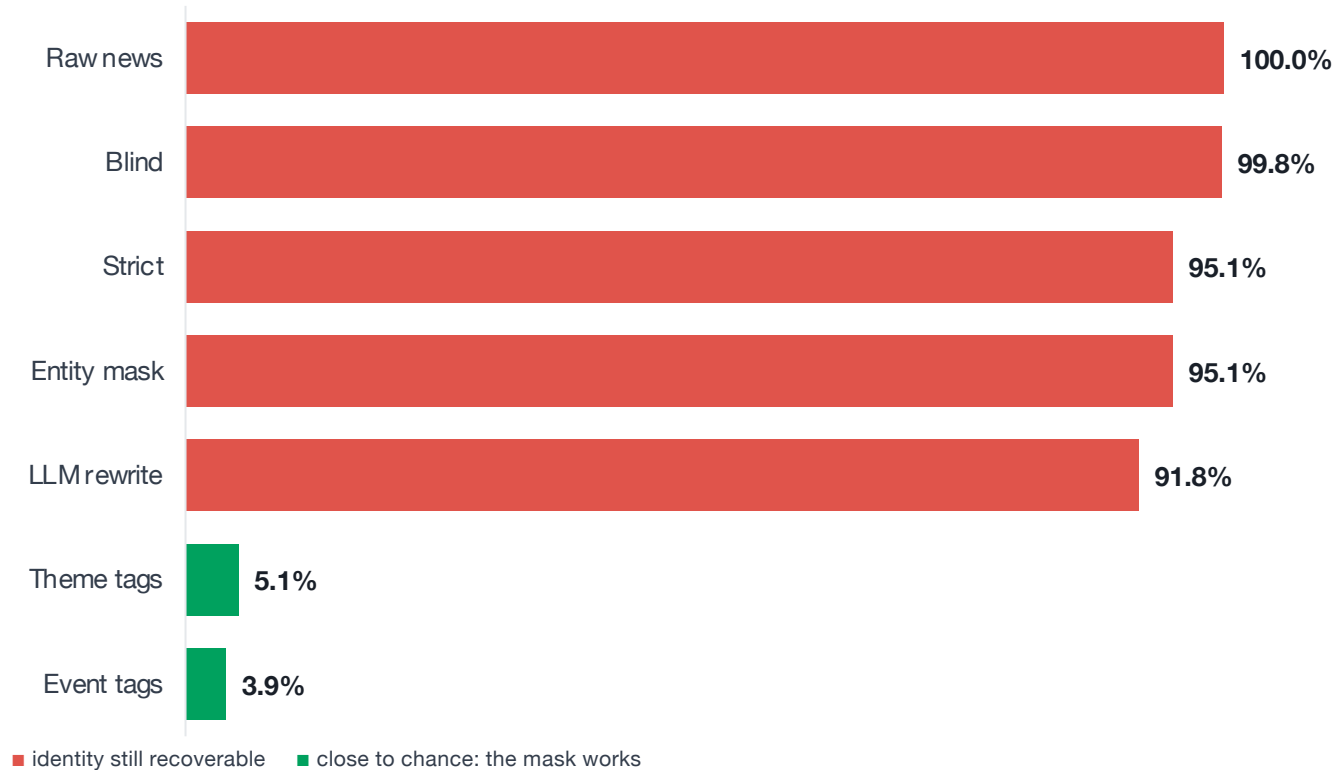
surviving identity cue anonymous label mask placeholder coarse tag

Redrawn from the paper's Figure 1. Chance of guessing the stock is 1.37%.

Masked records point straight back to the stock

Top-1 recovery of the source stock under the strongest of five fixed retrieval attacks (chance: 1.37%).

How often the attack finds the stock (lower is safer; 1.37% means the mask worked)



Source: AlphaLeak Fig. 1 and §4.2.

What it shows

- Blind keeps about nineteen entity-like tokens per record. Strict and Entity mask remove them, yet recovery stays at 0.951: the remaining wording carries the identity.
- It holds with a 17× larger candidate pool (0.987) and an open-set test with impostors (ROC-AUC 0.991).
- Theme and Event tags cut recovery to 2.8–3.7× chance: risk reduction, not certified blindness.

Hidden or useful, never both

Each point is one representation: how identifiable it is vs how much ranking value it adds over numbers alone.



Panel A. Strict and LLM rewrite are omitted: the paper reports their utility only graphically.

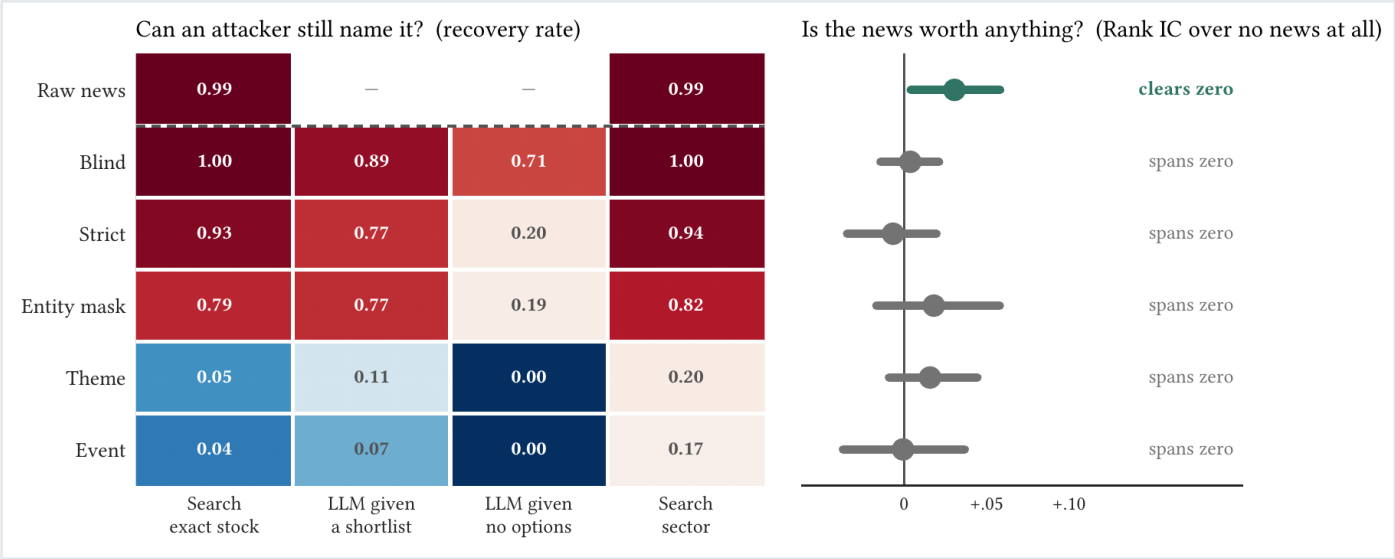
Source: AlphaLeak Figs. 1–3, Tables 1–2 and §4.3–4.6.

What it shows

- Rank IC measures whether the model ordered the stocks correctly; 0 means no better than ignoring the news, and higher is better.
- Only Raw news clears zero, +0.0305 [+0.004, +0.058], and it is fully identifiable.
- The gain is sector-level: Raw – Blind is +0.1292 across sector means but –0.0123 within sectors. Portfolio: +43 bp/month = +53 sector tilt – 10 within.
- Pooled across four LLMs, the effect is +0.0041 [–0.0084, +0.0167].

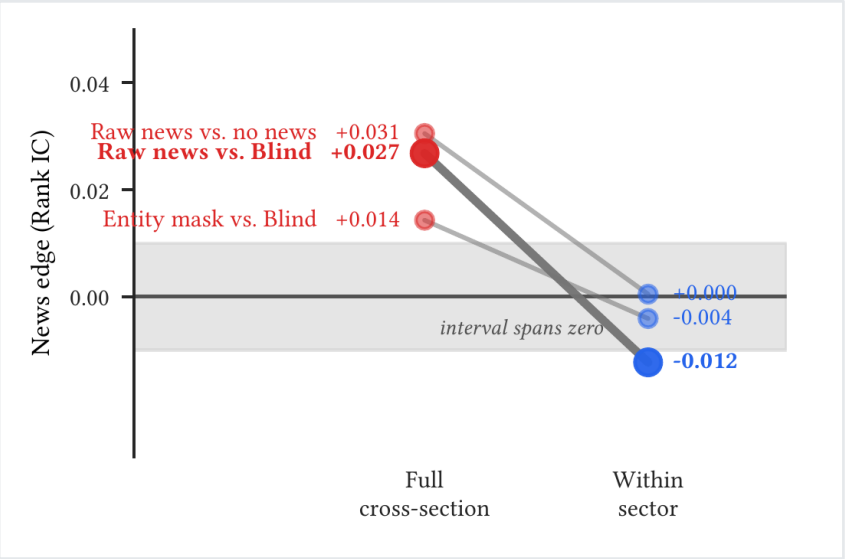
The audit scorecard, as published

Original figures from the submitted paper, shown as published. Numbers on earlier slides are read from its tables and text.



Paper Figure 2: Audit scorecard for the six representations

Left: recovery under four attacks; darker red is easier to re-identify. Right: Rank IC over No news with 95% intervals; only Raw news clears zero.



Paper Figure 3: The news edge is a sector tilt, not stock selection

Every Raw news contrast is positive across the full cross-section and falls to about zero or below once sectors are neutralized.

What held, and what the paper does not claim

The audit tests every released surface, not only the news text.

Checks that held

- Five attack families plus two LLM adversaries, including one allowed to answer “unknown.”
- The 200-stock replication keeps the same ordering (Strict 0.955, Event 0.025).
- Non-text leaks: the six numeric fields alone re-identify every record (1.000); 17-date score histories re-identify 73–90%.
- Random, same-sector and cross-sector placebo articles; four scoring models; prompt-order shuffles.

What the paper does not claim

- It shows benchmark-specific linkability, not that every model actually inferred the ticker.
- The scoring models postdate the 2023–2024 window, so memorization may inflate both recovery and utility.
- Small sample: 17 clusters and a minimum detectable effect of about 0.036; portfolio returns are gross of costs.
- Recommendation: “Masking should describe what was done; blindness should describe what survived an audit.”

Code (anonymous review repository): anonymous.4open.science/r/ICAIF-Mask-No-Blind